

1.1 Radix sort works by sorting numbers by digits.

Starting from least significant to most.

3 digit number, 3 passes

1.

432 $\rightarrow$ 2

235 $\rightarrow$ 5

129 $\rightarrow$ 9

543 $\rightarrow$ 3

602 $\rightarrow$ 2

764 $\rightarrow$ 4

230 $\rightarrow$ 0

987 $\rightarrow$ 7

564 $\rightarrow$ 4

345 $\rightarrow$ 5

Place based on ones digit

0 $\rightarrow$ 230

2: 432, 602

3: 543

4: 764, 564

5: 235, 345

7: 987

9: 129

Make list

~~250, 4~~

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Second Pass for 10's digit

$230 \rightarrow 3$, $432 \rightarrow 3$, $602 \rightarrow 0$, $543 \rightarrow 4$, $764 \rightarrow 6$
 $564 \rightarrow 6$, $235 \rightarrow 3$, $345 \rightarrow 4$, $987 \rightarrow 8$
 $129 \rightarrow 2$

Place Based on 10's

0: 602
2: 129
3: 230, 432, 235
4: 543, 345
6: 764, 564
8: 987

3rd Pass Sorting by Hundred

$602 \rightarrow 6$, $129 \rightarrow 1$, $230 \rightarrow 2$, $432 \rightarrow 4$, $235 \rightarrow 2$
 $543 \rightarrow 5$, $345 \rightarrow 3$, $764 \rightarrow 7$, $564 \rightarrow 5$, $987 \rightarrow 9$

1: 129
2: 230, 235
3: 345
4: 432
5: 543, 564
6: 602
7: 764
9: 987

Final list

121, 230, 235, 345, 432, 543, 564, 602, 764, 987

1.2. Comparison

Radix sort is more efficient for large numbers of integers because it is a non-comparative algorithm that sorts digit by digit achieving time of $O(nk)$.
nis number elements, k is number of digits.

Comparison sorts have typically $O(n \log n)$

Radix sort works for smaller integers too, but the overhead of multiple passes and bucket management might not be justified for small datasets.

2.1

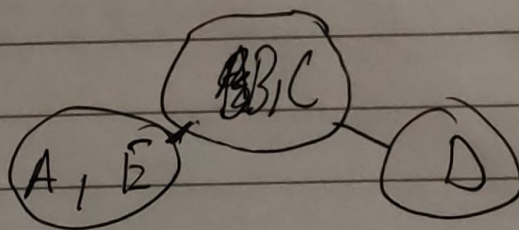
Munger's decision process.

1. Shipment size
2. Perishability
3. Alphabetical order

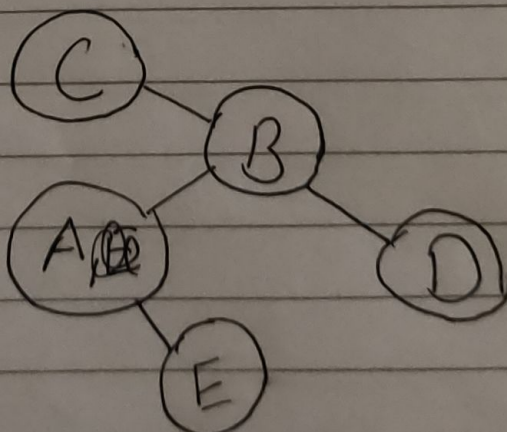
example setup

Shipment	size	Perishability
A	50	Yes
B	60	No
C	60	Yes
D	40	No
E	50	No

1. First thing to do would be comparison on sizes



2. Now comparison on perishability



3. If still groups, tiebreaker of Alphabet.
- Not needed in example

Usage of tree

1- From 60 branch C, then B

2- then B

3- From 50 branch A

4 then E

5 Finally D

C	B	A	E	D
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2.2

Total possible Permutations = $10!$

$\therefore$ total leaves = $10!$

min height is $\log_2(10!)$

$$h > \log_2(10!)$$

Using Stirling's approximation $n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$

$$10! \approx 3.6 \times 10^6$$

$$h > \log_2(3.6) + 6 \log_2 10$$

$$h > 21.79$$

Ans

min height = 22